

Math 110 – Homework 8 Answers

Math 110 – Fall 2026

Homework 8

Generalized eigenvectors and Jordan form

Suggested completion date: Friday, October 23

Unless otherwise indicated, all vector spaces are over the field $\mathbb{F}$. Problems referring to LADR are from *Linear Algebra Done Right*, 4th ed..

Problem 1. Let $D : \mathcal{P}_n(\mathbb{C}) \rightarrow \mathcal{P}_n(\mathbb{C})$ be differentiation.

- (a) Prove that D is nilpotent and determine its nilpotency index.
- (b) Determine the geometric and algebraic multiplicities of the eigenvalue 0.
- (c) Find a basis with respect to which the matrix of D is a single Jordan block.

Problem 2. Construct explicit matrices having the following spectral data.

- (a) Eigenvalue 2 has algebraic multiplicity 3 and geometric multiplicity 1; eigenvalue -1 has algebraic multiplicity 2 and geometric multiplicity 2.
- (b) The same algebraic multiplicities as in (a), but the geometric multiplicity of 2 is 2.

For each matrix, give its minimal polynomial.

Problem 3. Let

$$J = \begin{pmatrix} 1 & 1 & 0 \\ 0 & 1 & 1 \\ 0 & 0 & 1 \end{pmatrix}.$$

- (a) Compute $\text{null}(J - I)$, $\text{null}((J - I)^2)$, and $\text{null}((J - I)^3)$.
- (b) Determine the generalized eigenspace $G(1, J)$ and the geometric and algebraic multiplicities of 1.
- (c) Compute J^m for every positive integer m .

Problem 4. Let N be nilpotent with Jordan form consisting of one block of size 3 and one block of size 2.

- (a) Compute $\dim \text{null } N^k$ for $k = 0, 1, 2, 3, 4$.
- (b) Suppose instead that you are only told

$$\dim \text{null } N = 2, \quad \dim \text{null } N^2 = 4, \quad \dim \text{null } N^3 = 5.$$

Explain how these numbers force the same two block sizes.

Problem 5 (optional). Let $T \in \mathcal{L}(\mathbb{C}^6)$ have eigenvalues 1 and 2, and suppose

$$\dim \text{null}(T - I) = 1, \quad \dim \text{null}((T - I)^2) = 2, \quad \dim \text{null}((T - I)^3) = 2,$$

while

$$\begin{aligned}\dim \text{null}(T - 2I) &= 2, & \dim \text{null}((T - 2I)^2) &= 3, \\ \dim \text{null}((T - 2I)^3) &= 4, & \dim \text{null}((T - 2I)^4) &= 4.\end{aligned}$$

Determine the Jordan normal form of T , up to reordering of the blocks.

Problem 6 (optional). Suppose an operator has Jordan blocks

$$J_3(4), \quad J_2(4), \quad J_1(-1).$$

Determine:

- (a) its characteristic polynomial;
- (b) its minimal polynomial;
- (c) the geometric and algebraic multiplicities of each eigenvalue;
- (d) whether it is diagonalizable.

Problem 7 (optional). Prove that for a nilpotent operator N ,

$$\dim \text{null } N^k - \dim \text{null } N^{k-1}$$

is the number of Jordan blocks of size at least k .

Solutions

Problem 1

Solution notes

Problem 2

Solution notes

Problem 3

Solution notes

Problem 4

Solution notes

Problem 5**Solution notes**

Problem 6**Solution notes**

Problem 7**Solution notes**
